

Problem 291: Jammer Defense Network

Difficulty: Medium

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Problem Background

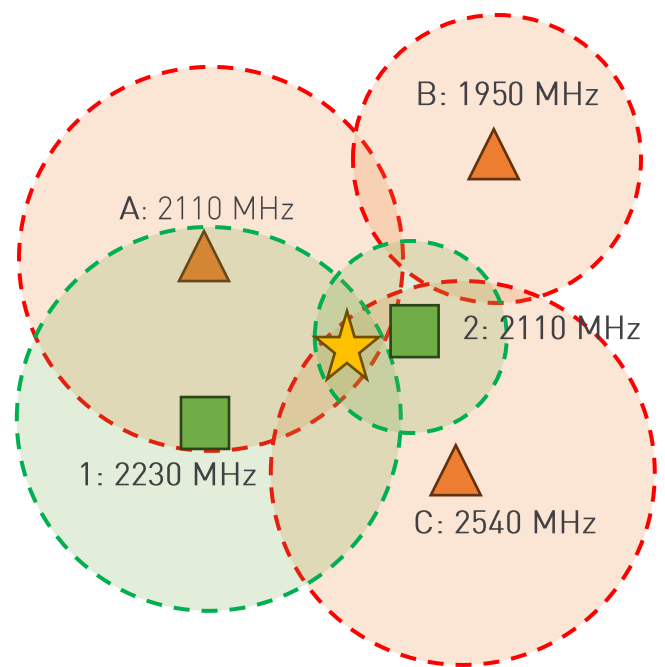
Lockheed Martin simulates wartime scenarios where a number of electronic warfare tactics are used to attack or defend Radio Frequencies (RF). This is known as jamming, the intentional interference of radio signals. RF jamming can be used defensively to block unwanted RF signals from interfering with strategic locations. Your team has been asked to help set up one of these simulations to assist with training teams from the United States Army.

Problem Description

To defend against RF interruptions, an Army team has deployed a series of jammers around their base of operations. These jammers are able to block out RF signals at a specific frequency and have a limited functional range, represented as a circle centered on the jammer with a given radius. The simulated “enemy,” another Army team, has constructed a number of RF transmitters in the area intended to be used to carry out attacks against the first team. These also work on specific frequencies and have an effective range. You are reviewing the placements of both team’s towers to determine if the first team is protected against the second team’s attacks.

The coordinates of the “defending” base operated by the first team are fixed at (0,0). You will be given the locations of the “enemy” RF transmitters and the “allied” RF jammers around that base, as well as the frequency on which each of those operates and their effective range. If the defending base is within range of a transmitter, the base will be exposed to attacks along that transmitter’s frequency... unless the base is also in range of a jammer operating at that same frequency. If the base is outside the range of a transmitter tower, that tower does not pose a threat.

In the diagram shown at right, the base is protected from any attacks from Transmitter A because Jammer 2 operates on the same frequency and is able to cover the base. Transmitter B is too far away



to be of any risk. The base is at risk of attacks from Transmitter C, however; even though the base is covered by Jammers 1 and 2, neither uses the same frequency as Transmitter C.

Given the information about the transmitters and jammers in the area, you'll need to determine if the Army's first team has succeeded in securing their base against the second team's attacks.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include:

- A line containing a positive integer, **N**, representing the number of towers around the base
- **N** lines, each containing information about a tower in use during the simulation. Each line will include the following information, separated by spaces:
 - An uppercase letter "J" or "T", indicating if the tower is an allied jammer ("J") or an enemy transmitter ("T").
 - The location of the tower, given in "X,Y" coordinate format. Both X and Y will be integers and will be separated by a comma.
 - A positive integer representing the range of the tower.
 - A positive integer representing the frequency of the tower in MHz.

```
2
5
T -5,3 10 2110
T 10,10 8 1950
T 6,-6 10 2540
J -5,-3 10 2230
J 3,1 5 2110
6
T -4,2 5 2360
J -6,12 15 2450
T 5,4 4 1770
J 3,-3 6 1890
T 7,7 10 2450
J 5,-8 10 2360
```

Sample Output

For each test case, your program must print a single line containing the word "SAFE" if the base is protected against any and all potential threats from enemy transmitters, or "UNSAFE" if the base is vulnerable to attack from at least one enemy transmitter.

```
UNSAFE
SAFE
```